

Thus, total time = $t_1 + 1 = 2.0$ s

5. Answer: D

In order for both spheres to be at rest at the same time, the total momentum of the spheres has to be zero.

Since $|MV| > |mv|$, the total momentum of the spheres cannot be zero. Since there is no net force acting on the system, the total momentum has to be conserved. Thus both spheres cannot be at rest at the same time.

6. Answer: C